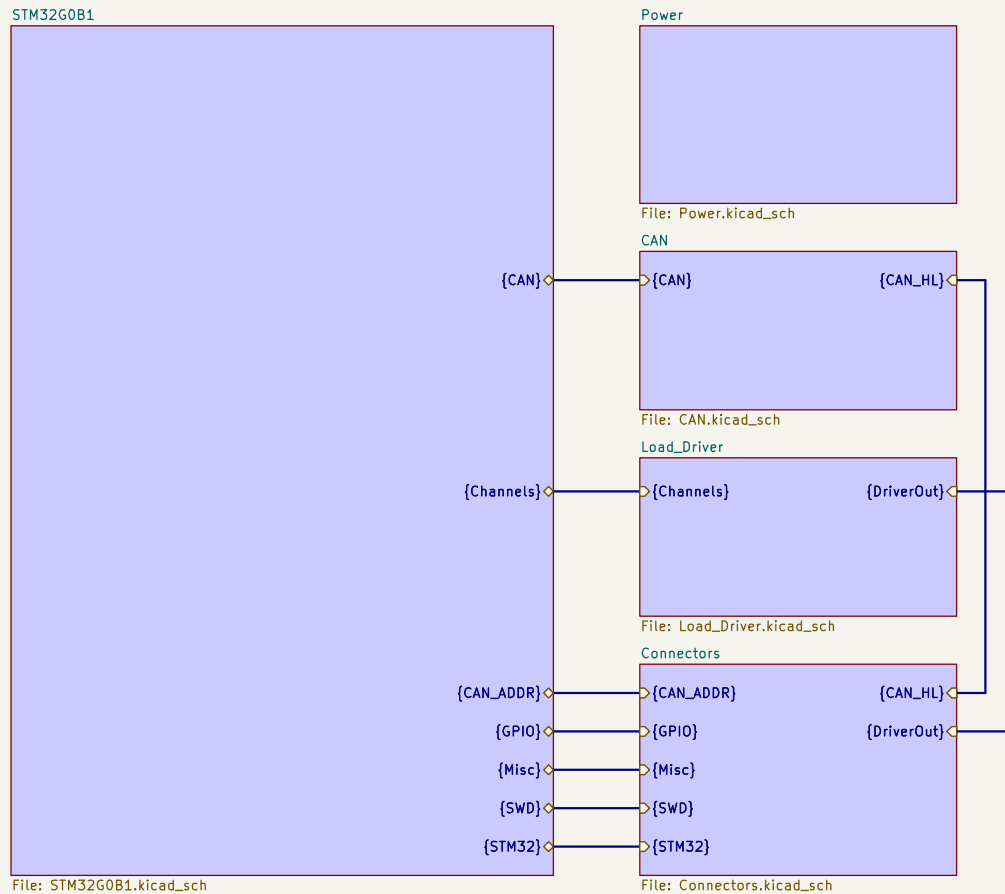


CAN IO Board for NJIT Solar Car

Rev A
06/2026



Author(s): Kostadin Madjovski, Arnav Revankar

NJIT Solar Car

Sheet: /

File: CANIO-Board.kicad_sch

Title: CAN IO Board

Size: USLetter Date: 2026-06-09

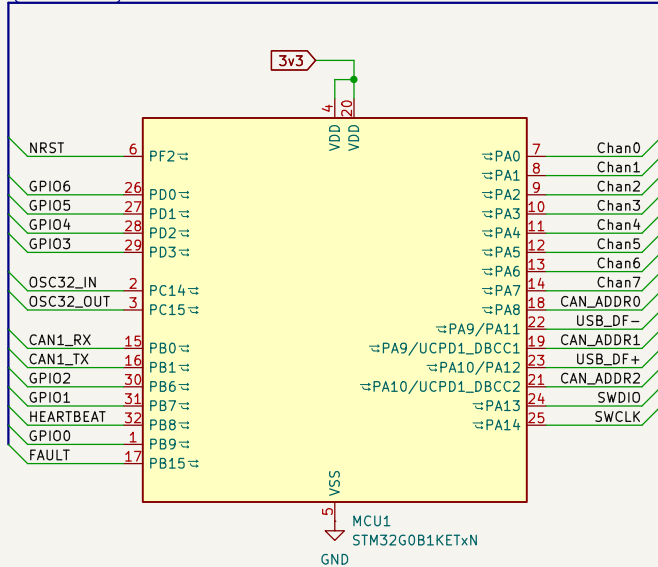
KiCad E.D.A. 9.0.2

Rev: A

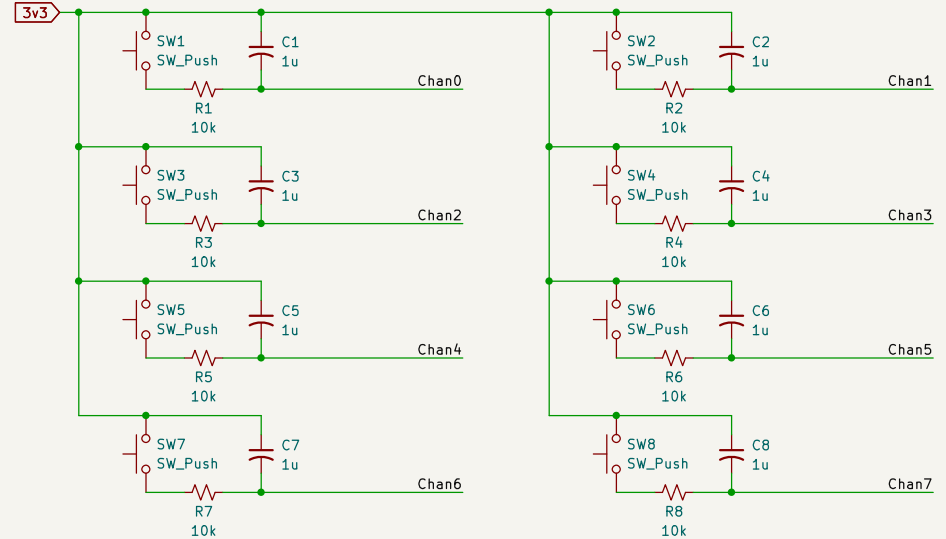
Id: 1/6

STM32G0B1

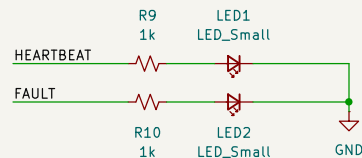
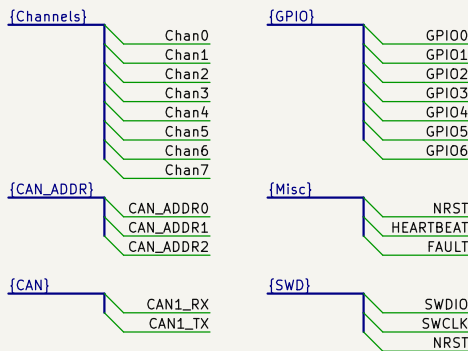
{STM32}



Channel Buttons (for debugging)



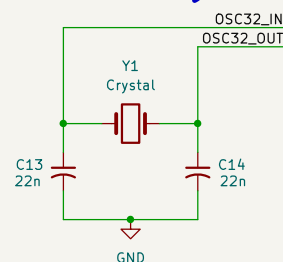
Labels and LEDs



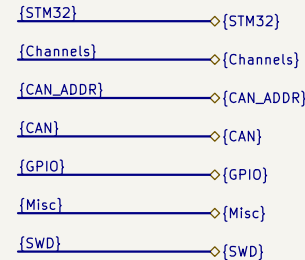
Design Notes

- Chan 0 through 7 are the gpio outputs which control the high current outputs 0 through 7
- All MCU pins are connected via buses, which connect to other hierarchical sheets
- GPIO[0..6] are to be used as needed later. Connector is TBD. They are collectively capable of I2C, SPI, and 2 timers (PWM)
- Channel buttons (used for debugging) are connected with RC low pass filter for button debouncing. Time constant is 10ms. Look at TI app note for info.
- No external pulldown on the channel inputs because the load driver chip comes with them internally. If using mosfets and relays, add external pulldowns

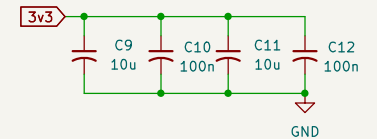
32.768kHz Crystal



Global/Hierarchical Labels



Decoupling Caps



Author(s): Arnav Revankar, Kostadin Madjovski

NJIT Solar Car

Sheet: /STM32G0B1/

File: STM32G0B1.kicad_sch

Title: CAN IO Board | STM32G0B1

Size: USLetter Date: 2026-06-09

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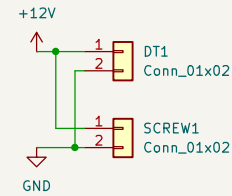
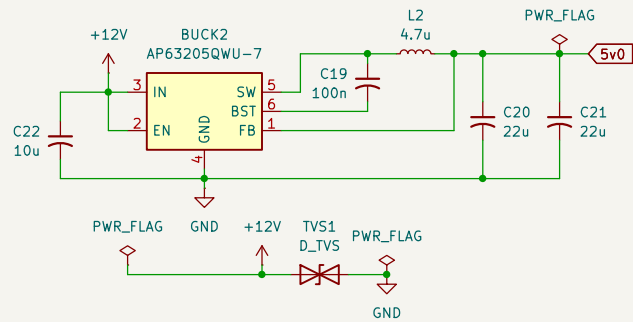
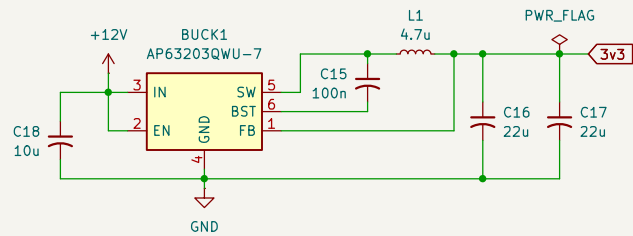


SOLAR CAR TEAM

Rev: A

Id: 2/6

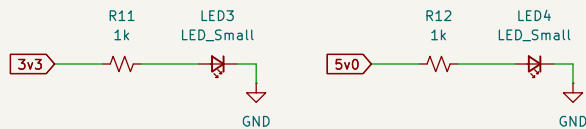
Buck Converters



Design Notes

- All parts of the schematic are from the manufacturer's datasheet.
- TVS diode since the 12V source is exposed
- HOWEVER: What is currently UNCLEAR is the 4.7uH and 2x22uF filtering setup at the output of each IC. Needs to be reviewed.
- Illustration at <https://www.desmos.com/calculator/au2a0t2rlf>

Power LEDs



Author(s): Kostadin Madjovski

NJIT Solar Car

Sheet: /Power/

File: Power.kicad_sch

Title: CAN IO Board | Power

Size: USLetter Date: 2026-06-09

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Rev: A

Id: 3/6

A



1

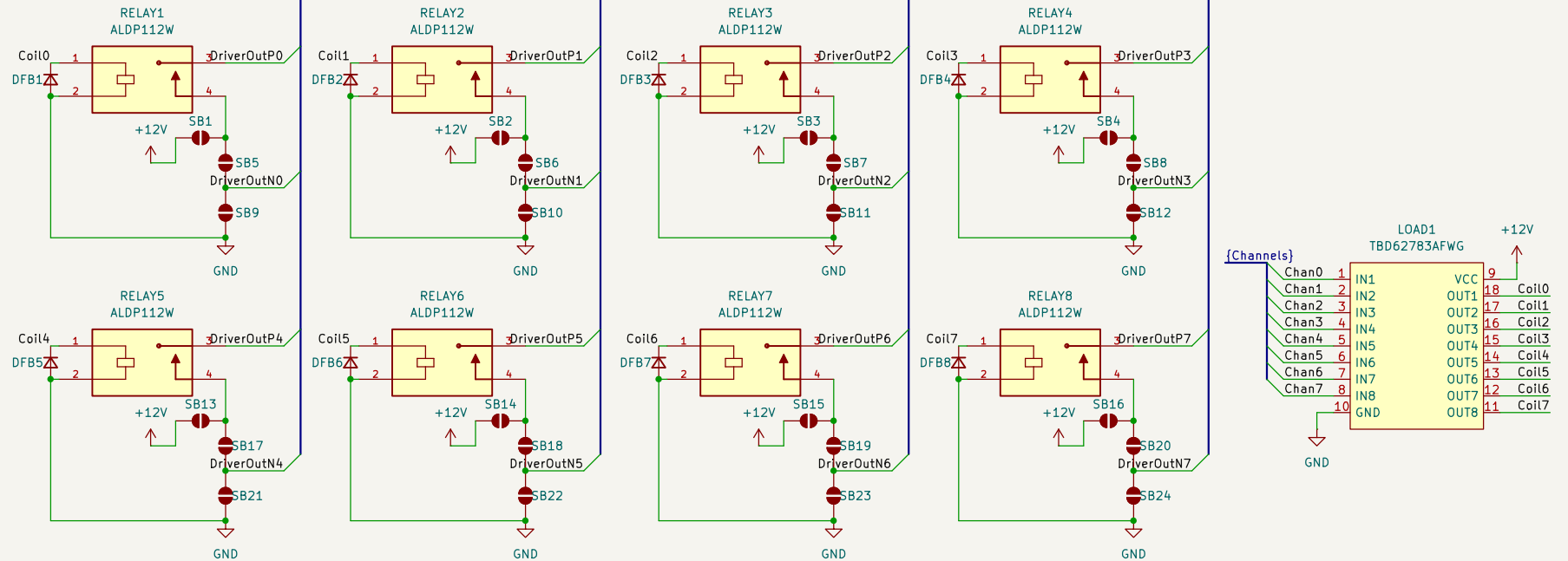
- 2

A



Id: 4/6

Load Driver High Side Switch and Relays



Design Notes

- The load driver IC is an 8 channel high side switch, that can supply upto 500mA per channel with Vcc upto 50V.
- The high side switch is used to power 8 relay coils, which each draw 12V 30mA (0.36W). The relays themselves are capable of supplying upto 15A constant.
- The solder bridges allow you to select from the WAGO connector output being 12V-GND, or just a connected/disconnected wire.
- For example, on the first relay, bridging SB1 and SB9 will put 12V and GND on the wago connector. On the other hand, bridging SB5 alone and leaving the other two open will simply connect the loop, DO NOT CONNECT IN ANY OTHER CONFIGURATION!!!
- Of all the high-current components, the solder bridge is the weak link. Thus, using the biggest available solder bridge pad, max continuous current will be like 3 amps probably (not determined yet)
- Because of the previous point, all fuses on this sheet will be 5A fuses.

Global/Hierarchical Labels

{Channels}  {Channels}

{DriverOut}  {DriverOut}

Author(s): Arnav Revankar
NJIT Solar Car

Sheet: /Load_Driver/
File: Load_Driver.kicad_sch

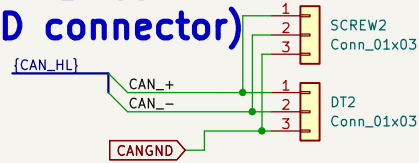
Title: CAN IO Board | Load Driver

Size: USLetter Date: 2026-06-09
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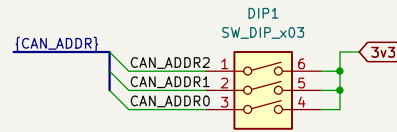


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CAN Lines (TD connector)



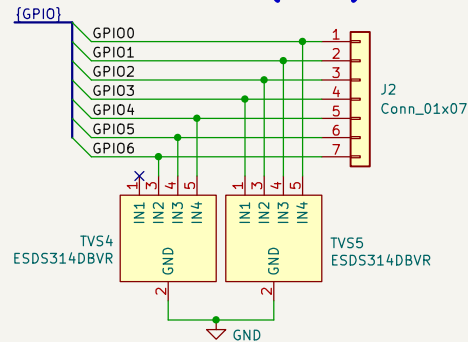
CAN Address Dip Switch



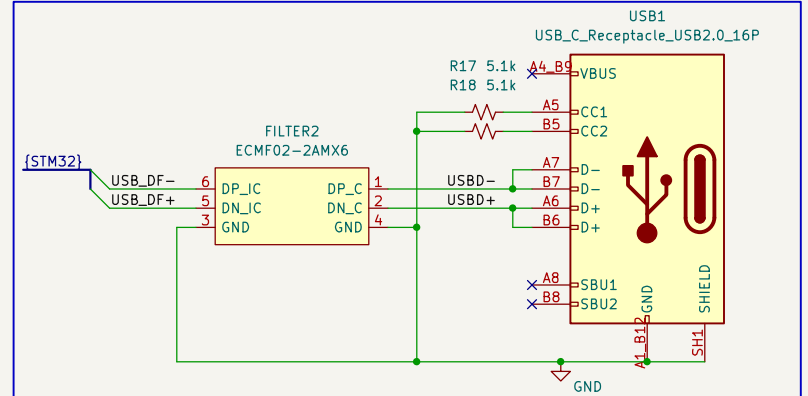
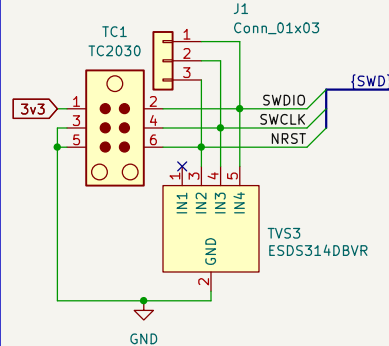
Design Notes

- The CANH and CANL lines are connected to the 3 pin TD connector. As of date the pinout is TBD, might be swapped.
- The CAN Address Dip Switch should control the least significant 3 bits of the board's CAN ID. This will allow upto 8 CAN ID variations, as per the reqs.
- The GPIOs are exposed, and thus have 4 channel unidirectional TVS diode arrays. RWM voltage is 3.6V.
- Since the SWD pins are exposed, there's the same 3.6V RWM diodes. But if the stlink programmer operates at 5V (gpios are 5V tolerant), will need to unsolder this IC.
- VBUS will NOT power the STM; external 12V is always required. Ground is shared.
- The USB DP, DM filter is MODIFIED from the custom library symbol. Restore if modified.
- NRST is active low with internal weak pullup.

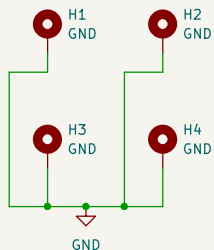
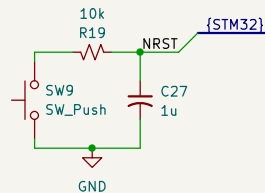
GPIO Connector (TBD)



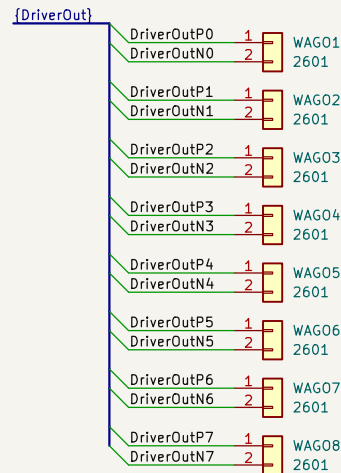
SWD



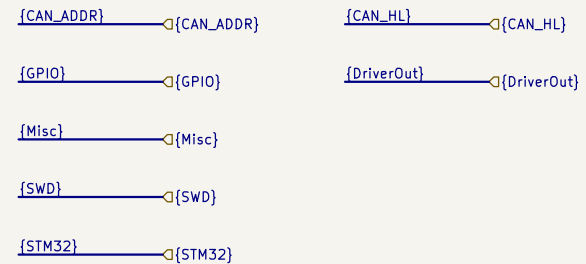
Reset and Mounting Holes



Channel Outputs



Global/Hierarchical Labels



Author(s): Arnav Revankar
NJIT Solar Car

Sheet: /Connectors/
File: Connectors.kicad_sch

Title: CAN IO Board | Connectors

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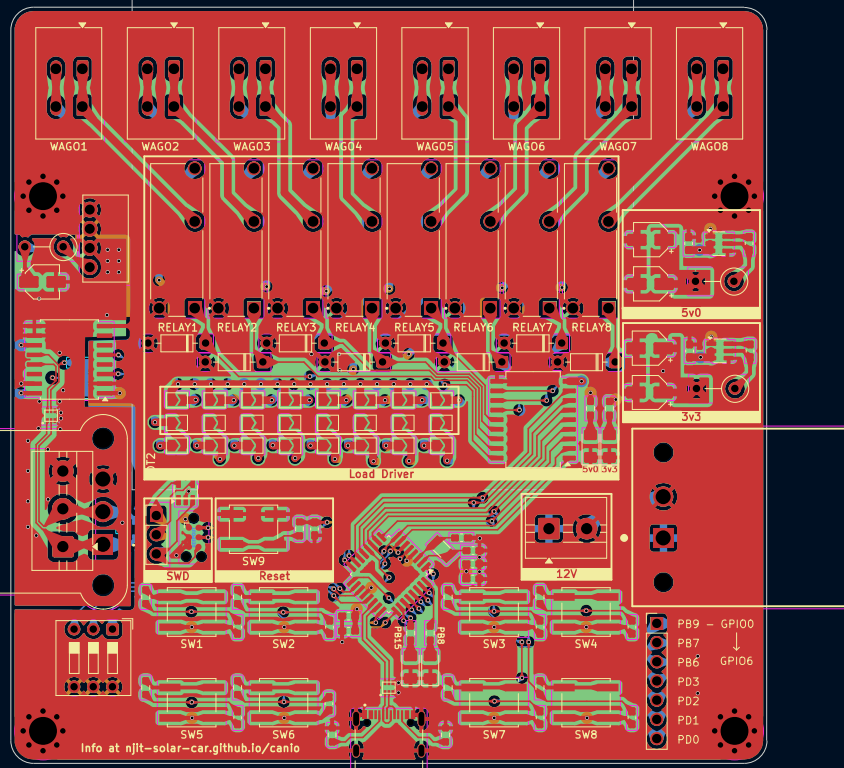


Rev: A
Id: 6/6

- Board is centered at (150, 115), and is 400mmx100mm.
- Corners have 3mm fillet. PTHs are 3.5mm (support M3 screws)
- Bottom PTHs are 4.25mm away from both edges. Top ones are 4.25mm away from the left and right edges, and 25mm away from the top and bottom edges. All from the center of board to center of PTH.
- MCU is 20mm below the center of the board, tilted 45 degrees for optimal routing.
- Allocated space for the high-current switching logic is centered at (450, 100), and is 600mmx300mm.
- When buttons are grouped left and right, they are positioned (20, 6) from the MCU.
- Cols carry only 30mA, so safe to go with thin traces
- Wavelength of CAN bus signal (1Mbps) in FR4 is ~150 meters. We don't really need impedance matching or a via plane, but adding it anyway. Using JLC's 04161H~7628 stub and TG135 core. KiCAD calculator gives can lines 0.16mm width and spacing both, I'm adding a via fence with 5mm spacing, 0.64mm away from the lines. This achieves 125Ohm impedance

NO Copper Pour on In1 under CANH/L1!!
Also, CANGND and CANVDD on top and bottom

Layer 4 – 12V
Layer 2 – GND
Layer 1 – GND



Sheet:
File: CANIO-Board.kicad_pcb

Size: USLetter	Date: 2026-06-15
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Rev: A
Id: 1/1

